

Julien Ferry

Postdoctoral Researcher at Polytechnique Montréal

Machine Learning Researcher | Trustworthy AI | Combinatorial Optimization

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Profile

Machine learning researcher working at the intersection of **machine learning** and **combinatorial optimization**, specializing in **trustworthy machine learning** for structured data. I use operations research and mathematical programming to develop accurate models with rigorous guarantees on robustness, fairness, interpretability, explainability, or privacy, with particular expertise in **tree ensembles** such as random forests and XGBoost. Beyond structured data, I also have experience with **deep learning**, particularly training CNNs for bias mitigation and robustness under distribution shift. I release my research as **open-source Python tools** and am especially interested in sensitive-data applications in healthcare, financial services, insurance, and risk assessment.

Recognition: Research nominated for Québec Science's Top 10 Scientific Discoveries of 2024; ICML 2024 oral presentation (top 1.5% of submissions); NeurIPS 2025 spotlight (top 3% of submissions); author or co-author of 16 peer-reviewed papers in leading machine-learning venues, including several as first author.

Supervision: Supervised 8 master students, currently co-supervising 2 PhD students.

Collaborations: I actively collaborate with researchers at various institutions in Montréal, elsewhere in Canada, and across Europe, including France, the Netherlands, Spain, Sweden, and Switzerland.

Academic service: Program committee member and reviewer for leading machine-learning conferences and journals, including ICML, NeurIPS, AACL, IJCAI, Neural Networks, TKDE, TMLR, and JMLR.

Languages: French (native); English (C1, TOEIC 955).

Education

2020–2023 | **Université Toulouse III Paul Sabatier / LAAS-CNRS, France**
PhD in Computer Science and Telecommunications
 Research on interpretability, fairness, and privacy through combinatorial optimization

2015–2020 | **INSA Toulouse, France**
Engineer's Degree in Electrical and Computer Engineering — Major: Distributed Systems and Big Data
 Ranked 1st in graduating class

Autumn 2017 | **The University of Texas at Austin**
Study semester abroad
 University Honors Fall Semester 2017

Professional Experience

Since Jan. 2024 | **Polytechnique Montréal — Postdoctoral Researcher, SCALE-AI Chair in Data-Driven Supply Chains**

- Developed training-data reconstruction attacks against tree ensembles to audit privacy risks in deployed models, including models protected by differential privacy
- Developed exact and efficient methods for generating counterfactual explanations for tree ensembles
- Developed methods to characterize, with formal guarantees, the trade-offs between predictive performance, fairness, interpretability, and privacy across several classes of interpretable and structured models
- Designed black-box model auditing frameworks with formal guarantees

2024–2025	BMDec Mobility Project — <i>External Research Collaborator</i> Developed privacy audit tools for statistics and models released from sensitive mobility data in a multidisciplinary project
November 2023	LAAS-CNRS, Toulouse — <i>Research Engineer, Operations Research and Constraints team</i> Applied combinatorial optimization to machine-learning methods following doctoral research in the same team
Oct. 2020–Oct. 2023	LAAS-CNRS / Université Toulouse III Paul Sabatier — <i>Doctoral Researcher</i> - Designed fair, interpretable classifiers using integer programming and sample-robust methods to improve fairness generalization - Investigated reconstruction of sensitive training attributes from model behavior - Characterized training data leakage for some classes of interpretable models
Feb.–July 2020	Thales Alenia Space, Toulouse — <i>Machine Learning Research Intern</i> Implemented a CNN bias-mitigation method and Python tools for bias analysis and correction; evaluated classification and semantic-segmentation use cases

Core Skills

- **Machine learning:** tree ensembles, optimal interpretable classifiers, counterfactual explanations, fairness evaluation and enforcement, model auditing, distributional robustness, characterization of Rashomon sets, privacy attacks and defenses, differential privacy
- **Optimization:** mixed-integer optimization, constraint programming, robust optimization, data structures, branch-and-bound algorithms, graph algorithms, metaheuristics
- **Programming and tools:** Python, scikit-learn, TensorFlow/Keras, OR-Tools, Gurobi, Git, Slurm, Numba

Selected Software and Research Projects

- **OCEAN:** Contributed to codebase refactoring and to the design and integration of new methods for counterfactual explanations in tree ensembles — code
- **HybridCORELS:** Python library for learning hybrid interpretable models that defer difficult instances to a black-box component — code
- **DRAFT:** Python implementation of a training-data reconstruction attack against random forests — code
- **FairCORELS / FairCORELSV2:** Python libraries for learning interpretable and statistically fair rule lists, including an exact integer-programming approach — code / V2
- **Rashomon-Explorer:** Python tools to characterize, with formal guarantees, the achievable trade-offs among accuracy, fairness, and interpretability within different hypothesis classes — code

Selected Publications

- **J. Ferry** et al. (2024). “Trained Random Forests Completely Reveal your Dataset.” *ICML*, oral presentation
- A. Khouna, **J. Ferry**, and T. Vidal (2025). “From Counterfactuals to Trees: Competitive Analysis of Model Extraction Attacks.” *NeurIPS*, spotlight
- A. Khouna, **J. Ferry**, and T. Vidal (2026). “Counterfactual Maps: What They Are and How to Find Them.” *NeurIPS*
- L. Langlade, **J. Ferry**, G. Laberge and T. Vidal (2025). “Fairness and Sparsity within Rashomon sets: Enumeration-Free Exploration and Characterization.” *AIES*
- **J. Ferry** et al. (2024). “Learning Hybrid Interpretable Models: Theory, Taxonomy, and Methods.” *TMLR*
- **J. Ferry** et al. (2023). “Improving Fairness Generalization Through a Sample-Robust Optimization Method.” *Machine Learning*

Full publication list: [Google Scholar](#).